

**Safety Data Sheet**

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**SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING****1.1 Product Identifier****Material Name** : GlycoShell Longlife Concentrate**Product Code** : 001B1272**1.2 Relevant identified uses of the substance or mixture and uses advised against****Product Use** : Antifreeze and coolant.**Uses Advised Against** : This product must not be used in applications other than those recommended in Section 1, without first seeking the advice of the supplier.**1.3 Details of the Supplier of the safety data sheet****Manufacturer/Supplier** : Shell UK Oil Products LimitedShell Centre  
London  
SE1 7NA  
United Kingdom**Telephone** : (+44) 08708500939**Email Contact for  
Safety Data Sheet** : If you have any enquiries about the content of this SDS please  
email lubricantSDS@shell.com**1.4 Emergency Telephone Number**

: +44-(0) 151-350-4595

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**SECTION 2. HAZARDS IDENTIFICATION****2.1 Classification of the substance or mixture**

| 1999/45/EC             |              |
|------------------------|--------------|
| Hazard Characteristics | R-phrases(s) |
| Xn: Harmful.;          | R22          |

**Classification triggering  
components** : Contains ethanediol. Contains bittering agent.

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### 2.2 Label Elements

#### Labeling according to Directive 1999/45/EC

EC Symbols : Xn Harmful.



EC Classification : Harmful.  
EC Risk Phrases : R22 Harmful if swallowed.  
EC Safety Phrases : S2 Keep out of the reach of children.  
S13 Keep away from food, drink and animal feeding stuffs.  
S46 If swallowed, seek medical advice immediately and show this container or label.

### 2.3 Other Hazards

**Health Hazards** : Slightly irritating to respiratory system.  
May cause moderate irritation to skin. Moderately irritating to eyes. Harmful if swallowed.  
May cause acidosis, cardiopulmonary and kidney effects.  
Ingestion may cause drowsiness and dizziness.  
Possibility of organ or organ system damage from prolonged exposure; see Chapter 11 for details. Target organ(s): Kidney.  
Lungs Cardiovascular system. Intentional abuse, misuse or other massive exposure may cause multiple organ damage and or death.

**Safety Hazards** : No specific hazards under normal use conditions.

**Environmental Hazards** : Not classified as dangerous for the environment.

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## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1 Substance

**Material Name** : Not applicable.

### 3.2 Mixtures

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**Mixture Description** : Mixture of monoethylene glycol and inhibitor package. Does not contain any amines, nitrates, phosphates, borates or silicates.

**Hazardous Components****Classification of components according to Regulation (EC) No 1272/2008**

| Chemical Name | CAS No.  | EC Number | REACH Registration No. | Conc. |
|---------------|----------|-----------|------------------------|-------|
| Ethanediol    | 107-21-1 | 203-473-3 | 01-2119456816-28       | %     |

| Chemical Name | Hazard Class & Category | Hazard Statement |
|---------------|-------------------------|------------------|
| Ethanediol    | Acute Tox., 4;          | H302;            |

**Classification of components according to 67/548/EEC**

| Chemical Name | CAS No.  | EC Number | REACH Registration No. | Symbol(s) | R-phrases | Conc.           |
|---------------|----------|-----------|------------------------|-----------|-----------|-----------------|
| Ethanediol    | 107-21-1 | 203-473-3 | 01-2119456816-28       | Xn        | R22       | 90.00 - 100.00% |

**Additional Information** : Refer to Ch 16 for full text of R- and H- phrases.

This mixture does not contain any REACH registered substances that are assessed to be a PBT or a vPvB.

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**SECTION 4. FIRST AID MEASURES****4.1 Description of First Aid Measures**

**General Information** : DO NOT DELAY. Keep victim calm. Obtain medical treatment immediately.

**Inhalation** : Remove to fresh air. If rapid recovery does not occur, transport to nearest medical facility for additional treatment.

**Skin Contact** : Remove contaminated clothing. Flush exposed area with water and follow by washing with soap if available. If persistent irritation occurs, obtain medical attention.

**Eye Contact** : Flush eye with copious quantities of water. If persistent irritation occurs, obtain medical attention.

**Ingestion** : DO NOT DELAY. If swallowed, do not induce vomiting; transport to nearest medical facility for additional treatment. If

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- vomiting occurs spontaneously, keep head below hips to prevent aspiration.
- Self-protection of the first aider** : When administering first aid, ensure that you are wearing the appropriate personal protective equipment according to the incident, injury and surroundings.
- 4.2 Most important symptoms and effects, both acute and delayed** : Kidney toxicity may be recognized by blood in the urine or increased or decreased urine flow. Other signs and symptoms can include nausea, vomiting, abdominal cramps, diarrhoea, lumbar pain shortly after ingestion, and possibly narcosis and death. High concentrations may cause central nervous system depression resulting in headaches, dizziness and nausea; continued exposure may result in unconsciousness and/or death.
- 4.3 Indication of any immediate medical attention and special treatment needed** : Notes to doctor/physician:  
IMMEDIATE TREATMENT IS EXTREMELY IMPORTANT!  
The preferred treatment is immediate transportation to a medical facility and use of appropriate treatment including possible administration of activated charcoal, gastric lavage and or gastric aspiration. If none of the above are immediately available and a delay of more than one hour is anticipated before such medical attention can be obtained, induction of vomiting may be appropriate using IPECAC syrup (Contraindicated if there are any signs of CNS depression). This should be considered on a case by case basis following specialist advice. Specific other treatments may include ethanol therapy, fomepizole, treatment of acidosis and haemodialysis. Seek specialist advice without delay.

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**SECTION 5. FIRE FIGHTING MEASURES**

Clear fire area of all non-emergency personnel.

- 5.1 Extinguishing Media** : Foam, water spray or fog. Dry chemical powder, carbon dioxide, sand or earth may be used for small fires only.
- Unsuitable Extinguishing Media** : Do not use water in a jet.
- 5.2 Special hazards arising from the substance or mixture** : Hazardous combustion products may include: A complex mixture of airborne solid and liquid particulates and gases (smoke). Carbon monoxide. Unidentified organic and inorganic compounds.
- 5.3 Advice for firefighters** : Proper protective equipment including chemical resistant gloves are to be worn; chemical resistant suit is indicated if large contact with spilled product is expected. Self-Contained

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Breathing Apparatus must be worn when approaching a fire in a confined space. Select fire fighter's clothing approved to relevant Standards (e.g. Europe: EN469).

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### SECTION 6. ACCIDENTAL RELEASE MEASURES

Avoid contact with spilled or released material. For guidance on selection of personal protective equipment see Chapter 8 of this Material Safety Data Sheet. Observe the relevant local and international regulations.

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|--------------------------------------------------------------------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>6.1 Personal Precautions, Protective Equipment and Emergency Procedures</b> | : | 6.1.1 For non emergency personnel: Avoid contact with skin and eyes.<br><br>6.1.2 For emergency responders: Avoid contact with skin and eyes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>6.2 Environmental Precautions</b>                                           | : | Use appropriate containment to avoid environmental contamination. Prevent from spreading or entering drains, ditches or rivers by using sand, earth, or other appropriate barriers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>6.3 Methods and Material for Containment and Cleaning Up</b>                | : | For large liquid spills (> 1 drum), transfer by mechanical means such as vacuum truck to a salvage tank for recovery or safe disposal. Do not flush away residues with water. Retain as contaminated waste. Allow residues to evaporate or soak up with an appropriate absorbent material and dispose of safely. Remove contaminated soil and dispose of safely.<br><br>For small liquid spills (< 1 drum), transfer by mechanical means to a labelled, sealable container for product recovery or safe disposal. Allow residues to evaporate or soak up with an appropriate absorbent material and dispose of safely. Remove contaminated soil and dispose of safely. |
| <b>Additional Advice</b>                                                       | : | Local authorities should be advised if significant spillages cannot be contained.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>6.4 Reference to other sections</b>                                         | : | For guidance on selection of personal protective equipment see Chapter 8 of this Material Safety Data Sheet. For guidance on disposal of spilled material see Chapter 13 of this Material Safety Data Sheet.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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### SECTION 7. HANDLING AND STORAGE

- |                            |   |                                                                                                                                                                                            |
|----------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Precautions</b> | : | Use local exhaust ventilation if there is risk of inhalation of vapours, mists or aerosols. Use the information in this data sheet as input to a risk assessment of local circumstances to |
|----------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Safety Data Sheet****7.1 Precautions for Safe Handling**

help determine appropriate controls for safe handling, storage and disposal of this material.

: Avoid prolonged or repeated contact with skin. Avoid inhaling vapour and/or mists. When handling product in drums, safety footwear should be worn and proper handling equipment should be used. Properly dispose of any contaminated rags or cleaning materials in order to prevent fires. Keep container tightly closed and in a cool, well-ventilated place. Use properly labelled and closeable containers.

**7.2 Conditions for safe storage, including any incompatibilities**

: Store at ambient temperature.

Refer to section 15 for any additional specific legislation covering the packaging and storage of this product. The storage of this product may be subject to the Control of Pollution (Oil Storage) (England) Regulations. Further guidance may be obtained from the local environmental agency office.

**Recommended Materials**

: For containers or container linings, use mild steel or high density polyethylene.

**Unsuitable Materials**

: Zinc. Avoid contact with galvanized materials.

**7.3 Specific end use(s)**

: Please refer to Ch16 and/or the annexes for the registered uses under REACH.

**Additional Information**

: Polyethylene containers should not be exposed to high temperatures because of possible risk of distortion. Exposure to this product should be reduced as low as reasonably practicable. Reference should be made to the Health and Safety Executive's publication "COSHH Essentials".

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**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION**

If the American Conference of Governmental Industrial Hygienists (ACGIH) value is provided on this document, it is provided for information only.

**8.1 Control Parameters****Occupational Exposure Limits**

| Material   | Source   | Type              | ppm | mg/m3    | Notation |
|------------|----------|-------------------|-----|----------|----------|
| Ethanediol | EH40 WEL | TWA(Particulate.) |     | 10 mg/m3 |          |

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|  |          |                        |        |           |                                   |
|--|----------|------------------------|--------|-----------|-----------------------------------|
|  | EH40 WEL | TWA(Vapor.)            | 20 ppm | 52 mg/m3  |                                   |
|  | EH40 WEL | STEL(Vapor.)           | 40 ppm | 104 mg/m3 |                                   |
|  | EH40 WEL | SKIN_DES(Particulate.) |        |           | Can be absorbed through the skin. |
|  | EH40 WEL | SKIN_DES(Vapor.)       |        |           | Can be absorbed through the skin. |
|  | ACGIH    | Ceiling(Aero sol.)     |        | 100 mg/m3 |                                   |

**Biological Exposure Index (BEI)**

No biological limit allocated.

**PNEC related information** : Data not available

**Monitoring Methods** : Monitoring of the concentration of substances in the breathing zone of workers or in the general workplace may be required to confirm compliance with an OEL and adequacy of exposure controls. For some substances biological monitoring may also be appropriate. Validated exposure measurement methods should be applied by a competent person and samples analysed by an accredited laboratory. Examples of sources of recommended exposure measurement methods are given below or contact the supplier. Further national methods may be available.

National Institute of Occupational Safety and Health (NIOSH),  
USA: Manual of Analytical Methods <http://www.cdc.gov/niosh/>

Occupational Safety and Health Administration (OSHA), USA:  
Sampling and Analytical Methods <http://www.osha.gov/>

Health and Safety Executive (HSE), UK: Methods for the  
Determination of Hazardous Substances

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<http://www.hse.gov.uk/>

Institut für Arbeitsschutz Deutschen Gesetzlichen  
Unfallversicherung (IFA), Germany.  
<http://www.dguv.de/inhalt/index.jsp>

L'Institut National de Recherche et de Sécurité, (INRS), France  
<http://www.inrs.fr/accueil>

### 8.2 Exposure Controls General Information

: The level of protection and types of controls necessary will vary depending upon potential exposure conditions. Select controls based on a risk assessment of local circumstances. Appropriate measures include: Adequate ventilation to control airborne concentrations. Where material is heated, sprayed or mist formed, there is greater potential for airborne concentrations to be generated.

Define procedures for safe handling and maintenance of controls. Educate and train workers in the hazards and control measures relevant to normal activities associated with this product. Ensure appropriate selection, testing and maintenance of equipment used to control exposure, e.g. personal protective equipment, local exhaust ventilation. Drain down system prior to equipment break-in or maintenance. Retain drain downs in sealed storage pending disposal or for subsequent recycle. Always observe good personal hygiene measures, such as washing hands after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants. Discard contaminated clothing and footwear that cannot be cleaned. Practice good housekeeping.

Do not ingest. If swallowed then seek immediate medical assistance.

### Occupational Exposure Controls

#### Personal Protective Equipment

: The provided information is made in consideration of the PPE directive (Council Directive 89/686/EEC) and the CEN European Committee for Standardisation (CEN) standards. Personal protective equipment (PPE) should meet recommended national standards. Check with PPE suppliers.

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- Eye Protection** : Wear safety glasses or full face shield if splashes are likely to occur. Approved to EU Standard EN166.
- Hand Protection** : Where hand contact with the product may occur the use of gloves approved to relevant standards (e.g. Europe: EN374, US: F739) made from the following materials may provide suitable chemical protection: PVC, neoprene or nitrile rubber gloves. Suitability and durability of a glove is dependent on usage, e.g. frequency and duration of contact, chemical resistance of glove material, dexterity. Always seek advice from glove suppliers. Contaminated gloves should be replaced. Personal hygiene is a key element of effective hand care. Gloves must only be worn on clean hands. After using gloves, hands should be washed and dried thoroughly. Application of a non-perfumed moisturizer is recommended. For continuous contact we recommend gloves with breakthrough time of more than 240 minutes with preference for > 480 minutes where suitable gloves can be identified. For short-term/splash protection we recommend the same, but recognise that suitable gloves offering this level of protection may not be available and in this case a lower breakthrough time may be acceptable so long as appropriate maintenance and replacement regimes are followed. Glove thickness is not a good predictor of glove resistance to a chemical as it is dependent on the exact composition of the glove material.
- Body protection** : Skin protection not ordinarily required beyond standard issue work clothes.
- Respiratory Protection** : No respiratory protection is ordinarily required under normal conditions of use. In accordance with good industrial hygiene practices, precautions should be taken to avoid breathing of material. If engineering controls do not maintain airborne concentrations to a level which is adequate to protect worker health, select respiratory protection equipment suitable for the specific conditions of use and meeting relevant legislation. Check with respiratory protective equipment suppliers. Where air-filtering respirators are suitable, select an appropriate combination of mask and filter. Select a filter suitable for combined particulate/organic gases and vapours [boiling point >65 °C (149 °F)] meeting EN14387.
- Thermal Hazards** : Not applicable.

**Environmental Exposure Controls**

- Environmental exposure control measures** : Minimise release to the environment. An environmental assessment must be made to ensure compliance with local

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environmental legislation. Take appropriate measures to fulfil the requirements of relevant environmental protection legislation. Avoid contamination of the environment by following advice given in Chapter 6. If necessary, prevent undissolved material from being discharged to waste water. Waste water should be treated in a municipal or industrial waste water treatment plant before discharge to surface water.

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**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES****9.1 Information on basic physical and chemical properties**

|                                                 |                                                            |
|-------------------------------------------------|------------------------------------------------------------|
| Appearance                                      | : Clear. Liquid at room temperature.                       |
| Odour                                           | : Characteristic.                                          |
| Odour threshold                                 | : Data not available                                       |
| pH                                              | : Typical 8.2 - 9 at 20 °C / 68 °F                         |
| Initial Boiling Point and Boiling Range         | : > 160 °C / 320 °F                                        |
| Freezing Point                                  | : Typical -18 °C / 0 °F                                    |
| Flash point                                     | : > 120 °C / 248 °F (COC)                                  |
| Upper / lower Flammability or Explosion limits  | : 3 - 15 %(V)                                              |
| Auto-ignition temperature                       | : ca. 400 °C / 752 °F                                      |
| Vapour pressure                                 | : Typical 2 hPa                                            |
| Relative Density                                | : Typical 1.12 at 20 °C / 68 °F                            |
| Density                                         | : Typical 1,122 - 1,125 kg/m <sup>3</sup> at 20 °C / 68 °F |
| Water solubility                                | : Completely Soluble                                       |
| Solubility in other solvents                    | : Data not available                                       |
| n-octanol/water partition coefficient (log Pow) | : Data not available                                       |
| Dynamic viscosity                               | : Data not available                                       |
| Kinematic viscosity                             | : Typical 22 - 26 mm <sup>2</sup> /s at 20 °C / 68 °F      |
| Vapour density (air=1)                          | : Data not available                                       |
| Evaporation rate (nBuAc=1)                      | : Data not available                                       |
| Decomposition                                   | : Data not available                                       |
| Temperature                                     |                                                            |
| Flammability                                    | : Data not available                                       |
| Oxidizing Properties                            | : Data not available                                       |
| Explosive Properties                            | : Not classified                                           |

**9.2 Other Information**

|                         |                                                             |
|-------------------------|-------------------------------------------------------------|
| Electrical conductivity | : This material is not expected to be a static accumulator. |
|-------------------------|-------------------------------------------------------------|

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Other Information : not a VOC  
Volatile organic compound : 0 %

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**SECTION 10. STABILITY AND REACTIVITY**

**10.1 Reactivity** : The product does not pose any further reactivity hazards in addition to those listed in the following sub-paragraph.  
**10.2 Chemical stability** : No hazardous reaction is expected when handled and stored according to provisions.  
**10.3 Possibility of Hazardous Reactions** :  
Reacts with strong oxidising agents.  
**10.4 Conditions to Avoid** : Extremes of temperature and direct sunlight.  
**10.5 Incompatible Materials** : Strong oxidising agents.  
**10.6 Hazardous Decomposition Products** : Hazardous decomposition products are not expected to form during normal storage.

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**SECTION 11. TOXICOLOGICAL INFORMATION****11.1 Information on Toxicological effects**

**Basis for Assessment** : Information given is based on data on the components and the toxicology of similar products.  
Unless indicated otherwise, the data presented is representative of the product as a whole, rather than for individual component(s).

**Likely Routes of Exposure** : Exposure may occur via inhalation, ingestion, skin absorption, skin or eye contact, and accidental ingestion.  
**Acute Oral Toxicity** : Expected to be of low toxicity: LD50 > 5000 mg/kg , Rat  
**Acute Dermal Toxicity** : Expected to be of low toxicity: LD50 > 5000 mg/kg , Rabbit  
**Acute Inhalation Toxicity** : Low toxicity: LC50 >5 mg/l / 4 h, Rat  
**Skin corrosion/irritation** : Expected to be slightly irritating.  
**Serious eye damage/irritation** : Expected to be slightly irritating.  
**Respiratory Irritation** : Inhalation of vapours or mists may cause irritation.  
**Respiratory or skin sensitisation** : For respiratory and skin sensitisation: Not expected to be a sensitiser.  
**Aspiration Hazard** : Not considered an aspiration hazard.

**Germ cell mutagenicity** : Not considered a mutagenic hazard.

**Carcinogenicity** : Not expected to be carcinogenic.

|            |                                                |
|------------|------------------------------------------------|
| Ethanediol | : GHS / CLP: No carcinogenicity classification |
|------------|------------------------------------------------|

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**Reproductive and Developmental Toxicity** : Causes foetotoxicity in animals; considered to be secondary to maternal toxicity.

**Summary on evaluation of the CMR properties**

**Carcinogenicity** : This product does not meet the criteria for classification in categories 1A/1B.,

**Mutagenicity** : This product does not meet the criteria for classification in categories 1A/1B.

**Reproductive Toxicity (fertility)** : This product does not meet the criteria for classification in categories 1A/1B.

**Specific target organ toxicity - single exposure** : Ingestion may cause drowsiness and dizziness. Inhalation of vapours or mists may cause irritation to the respiratory system.

**Specific target organ toxicity - repeated exposure** : Kidney: can cause kidney damage.

**Additional Information** : Classifications by other authorities under varying regulatory frameworks may exist.

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**SECTION 12. ECOLOGICAL INFORMATION**

**Basis for Assessment** : Ecotoxicological data have not been determined specifically for this product. Information given is based on a knowledge of the components and the ecotoxicology of similar products. Unless indicated otherwise, the data presented is representative of the product as a whole, rather than for individual component(s).

**12.1 Toxicity**  
**Acute Toxicity** : Expected to be practically non toxic: LC/EC/IC50 > 100 mg/l (to aquatic organisms)

**12.2 Persistence and degradability** : Readily biodegradable.

**12.3 Bioaccumulative Potential** : Not expected to bioaccumulate significantly.

**12.4 Mobility in Soil** : Liquid under most environmental conditions. If product enters soil, it will be highly mobile and may contaminate groundwater. Dissolves in water.

**12.5 Result of PBT and vPvB assesment** : This mixture does not contain any REACH registered substances that are assessed to be a PBT or a vPvB.

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**12.6 Other Adverse Effects** : Not expected to have ozone depletion potential, photochemical ozone creation potential or global warming potential.

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### SECTION 13. DISPOSAL CONSIDERATIONS

#### 13.1 Waste Treatment Methods

**Material Disposal** : Recover or recycle if possible. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste classification and disposal methods in compliance with applicable regulations. Do not dispose into the environment, in drains or in water courses.

**Container Disposal** : Dispose in accordance with prevailing regulations, preferably to a recognised collector or contractor. The competence of the collector or contractor should be established beforehand.

**Local Legislation** : Disposal should be in accordance with applicable regional, national, and local laws and regulations.  
EU Waste Disposal Code (EWC): 16 01 14 antifreeze fluids containing dangerous substances. Classification of waste is always the responsibility of the end user.

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### SECTION 14. TRANSPORT INFORMATION

#### Land transport (ADR/RID):

##### ADR

This product is not classified as dangerous for this mode of transport. Therefore 14.1 UN Number, 14.2 UN Proper Shipping name, 14.3 Transport hazard class(es), 14.4 Packing group, 14.5 Environmental hazards, 14.6 Special precautions for user do not apply.

##### RID

This product is not classified as dangerous for this mode of transport. Therefore 14.1 UN Number, 14.2 UN Proper Shipping name, 14.3 Transport hazard class(es), 14.4 Packing group, 14.5 Environmental hazards, 14.6 Special precautions for user do not apply.

#### Inland waterways transport (ADN):

This product is not classified as dangerous for this mode of transport. Therefore 14.1 UN Number, 14.2 UN Proper Shipping name, 14.3 Transport hazard class(es), 14.4 Packing group, 14.5 Environmental hazards, 14.6 Special precautions for user do not apply.

#### Sea transport (IMDG Code):

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This product is not classified as dangerous for this mode of transport. Therefore 14.1 UN Number, 14.2 UN Proper Shipping name, 14.3 Transport hazard class(es), 14.4 Packing group, 14.5 Environmental hazards, 14.6 Special precautions for user do not apply.

### Air transport (IATA):

This product is not classified as dangerous for this mode of transport. Therefore 14.1 UN Number, 14.2 UN Proper Shipping name, 14.3 Transport hazard class(es), 14.4 Packing group, 14.5 Environmental hazards, 14.6 Special precautions for user do not apply.

### 14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Pollution Category : Not applicable.  
Ship Type : Not applicable.  
Product Name : Not applicable.  
Special Precaution : Not applicable.

**Additional Information** : MARPOL Annex 1 rules apply for bulk shipments by sea.

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## SECTION 15. REGULATORY INFORMATION

The regulatory information is not intended to be comprehensive. Other regulations may apply to this material.

### 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

#### Other regulatory Information

**Authorisations and/or restrictions on use** : Product is not subject to Authorisation under REACH.

**Recommended Restrictions on Use (Advice Against)** : This product must not be used in applications other than those recommended in Section 1, without first seeking the advice of the supplier.

#### Chemical Inventory Status

EINECS : All components listed or polymer exempt.  
TSCA : All components listed.

**Other Information** : Environmental Protection Act 1990 (as amended).  
Health and Safety at Work etc. Act 1974.

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Consumers Protection Act 1987.  
Pollution Prevention and Control Act 1999.  
Environment Act 1995.  
Factories Act 1961.  
The Carriage of Dangerous Goods and Use of Transportable Pressure Equipment (Amendment) Regulations 2011.  
Chemicals (Hazard Information and Packaging for Supply) Regulations 2009.  
Control of Substances Hazardous to Health Regulations 2002 (as amended).  
Merchant Shipping (Dangerous Goods and Marine Pollutants) Regulations 1997.  
Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 1995 (as amended).  
Personal Protective Equipment Regulations 2002.  
Personal Protective Equipment at Work Regulations 1992.  
Hazardous Waste (England and Wales) Regulations 2005(as amended).  
Control of Major Accident Hazards Regulations 1999 (as amended).  
Renewable Transport Fuel Obligations Order 2007 (as amended).  
Energy Act 2011.  
Environmental Permitting (England and Wales) Regulations 2010 (as amended).  
Waste (England and Wales) Regulations 2011 (as amended).  
Planning (Hazardous Substances) Act 1990 and associated regulations.  
The Environmental Protection (Controls on Ozone-Depleting Substances) Regulations 2011.

**15.2 Chemical Safety Assessment** : No Chemical Safety Assessment has been carried out for this substance/mixture by the supplier.

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**SECTION 16. OTHER INFORMATION****R-phrases(s)**

R22 Harmful if swallowed.

**CLP Hazard Statements**

H302 Harmful if swallowed.

**Additional Information** : This product is classified as R22/H302 Harmful if swallowed.  
The same control advice applies to all uses of this product and

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is included in Section 8 of the SDS. An exposure scenario is not presented.

**Other Information****Abbreviations and Acronyms**

: Acute Tox. = Acute toxicity  
Asp. Tox. = Aspiration hazard  
Aquatic Acute = Acute hazards to the aquatic environment  
Aquatic Chronic = Hazardous to the aquatic environment - Long-term Hazard  
Eye Dam. = Serious eye damage/eye irritation  
Flam. Liq. = Flammable liquids  
Skin Corr. = Skin corrosion/irritation  
Skin Sens. = Skin sensitizer  
STOT SE = Specific target organ toxicity - single exposure  
STOT RE = Specific target organ toxicity - repeated exposure

The standard abbreviations and acronyms used in this document can be looked up in reference literature (e.g. scientific dictionaries) and/or websites.

ACGIH = American Conference of Governmental Industrial Hygienists  
ADR = European Agreement concerning the International Carriage of Dangerous Goods by Road  
AICS = Australian Inventory of Chemical Substances  
ASTM = American Society for Testing and Materials  
BEL = Biological exposure limits  
BTEX = Benzene, Toluene, Ethylbenzene, Xylenes  
CAS = Chemical Abstracts Service  
CEFIC = European Chemical Industry Council  
CLP = Classification Packaging and Labelling  
COC = Cleveland Open-Cup  
DIN = Deutsches Institut für Normung  
DMEL = Derived Minimal Effect Level  
DNEL = Derived No Effect Level  
DSL = Canada Domestic Substance List  
EC = European Commission  
EC50 = Effective Concentration fifty  
ECETOC = European Center on Ecotoxicology and Toxicology Of Chemicals  
ECHA = European Chemicals Agency  
EINECS = The European Inventory of Existing Commercial Chemical Substances  
EL50 = Effective Loading fifty

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ENCS = Japanese Existing and New Chemical Substances Inventory  
EWC = European Waste Code  
GHS = Globally Harmonised System of Classification and Labelling of Chemicals  
IARC = International Agency for Research on Cancer  
IATA = International Air Transport Association  
IC50 = Inhibitory Concentration fifty  
IL50 = Inhibitory Level fifty  
IMDG = International Maritime Dangerous Goods  
INV = Chinese Chemicals Inventory  
IP346 = Institute of Petroleum test method N° 346 for the determination of polycyclic aromatics DMSO-extractables  
KECI = Korea Existing Chemicals Inventory  
LC50 = Lethal Concentration fifty  
LD50 = Lethal Dose fifty per cent.  
LL/EL/IL = Lethal Loading/Effective Loading/Inhibitory loading  
LL50 = Lethal Loading fifty  
MARPOL = International Convention for the Prevention of Pollution From Ships  
NOEC/NOEL = No Observed Effect Concentration / No Observed Effect Level  
OE\_HP V = Occupational Exposure - High Production Volume  
PBT = Persistent, Bioaccumulative and Toxic  
PICCS = Philippine Inventory of Chemicals and Chemical Substances  
PNEC = Predicted No Effect Concentration  
REACH = Registration Evaluation And Authorisation Of Chemicals  
RID = Regulations Relating to International Carriage of Dangerous Goods by Rail  
SKIN\_DES = Skin Designation  
STEL = Short term exposure limit  
TRA = Targeted Risk Assessment  
TSCA = US Toxic Substances Control Act  
TWA = Time-Weighted Average  
vPvB = very Persistent and very Bioaccumulative

**SDS Distribution** : The information in this document should be made available to all who may handle the product.  
**SDS Version Number** : 3.0  
**SDS Effective Date** : 03.12.2012

## Safety Data Sheet

|                       |   |                                                                                                                                                                                                                                                               |
|-----------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SDS Revisions</b>  | : | A vertical bar ( ) in the left margin indicates an amendment from the previous version.                                                                                                                                                                       |
| <b>SDS Regulation</b> | : | Regulation 1907/2006/EC as amended by Regulation (EU) 453/2010                                                                                                                                                                                                |
| <b>Disclaimer</b>     | : | This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product. |